MARKET POTENTIAL & APPLICABILITY ELSEWHERE

AURORA (AI-Powered Engineering Design Optimization, Rule Compliance & 3D–BOM Synchronization Platform)

1. Executive Summary

AURORA is an AI-powered Engineering Design Intelligence Platform developed to automate engineering design compliance, optimize design decisions, synchronize CAD models with Bills of Material (BOM), and predict downstream engineering impacts before production.

The solution directly addresses one of the largest causes of schedule delays and cost overruns in engineering-intensive industries—the manual validation of engineering designs against standards, regulations and product configurations.

Although developed for the Indian Defence Shipbuilding ecosystem under the iDEX Drishti Challenge, the platform has applications across every industry where complex engineered products are designed, manufactured, maintained or certified.

Unlike traditional CAD or PLM software, AURORA functions as an Engineering Intelligence Layer sitting above existing engineering systems, providing continuous compliance validation, intelligent recommendations, automated traceability and change impact analysis.

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2. Industry Overview

Modern engineering organizations manage thousands of interconnected design components spread across:

• CAD Models

• Bills of Material

• Engineering Change Notices

• Design Standards

• Regulatory Documents

• Supplier Specifications

• Manufacturing Drawings

These artefacts are generally maintained across disconnected software systems, resulting in:

- Manual verification
- Human errors
- Design inconsistencies
- Late-stage engineering changes
- Manufacturing delays
- Increased lifecycle costs

MDL's challenge statement itself highlights that approximately **25% of engineering changes occur after contract design finalization**, resulting in downstream impacts across hull, propulsion, weapons and other ship systems.

AURORA addresses these issues through AI-assisted design validation, rule-based compliance checking, engineering knowledge graphs, and automated synchronization between 3D CAD models and Bills of Material.

3. Problem Statement

Organizations across engineering industries face common challenges:

Current Challenge	Business Impact
Manual rule checking	Slow engineering reviews
CAD and BOM mismatch	Manufacturing errors
Engineering changes	Cost escalation
Design revisions	Project delays
Regulatory compliance	Certification risks
Limited traceability	Difficult audits
Cross-disciplinary coordination	High rework

These challenges are universal across industries that produce high-value engineered products.

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4. Market Opportunity

The global market for Engineering Design Intelligence is driven by increasing digital transformation, Model-Based Systems Engineering (MBSE), Digital Twins, AI-assisted engineering, and Product Lifecycle Management (PLM).

Key market drivers include:

- AI in Engineering
- Industry 4.0
- Smart Manufacturing
- Defence Modernization
- Digital Twins
- Smart Factories
- Engineering Automation

The growing adoption of AI within engineering workflows creates a significant opportunity for intelligent design validation platforms such as AURORA.



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5. Serviceable Obtainable Market

Initial commercialization focuses on India.

Potential customers include:

Defence

- Mazagon Dock Shipbuilders Limited
- Garden Reach Shipbuilders & Engineers
- Goa Shipyard Limited

- Cochin Shipyard
- Hindustan Shipyard

Aerospace

- HAL
- DRDO Labs
- ISRO Centers

Manufacturing

- L&T Defence
- Tata Advanced Systems
- Bharat Forge
- BHEL

Projected obtainable market over five years:

USD 80–150 Million

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6. Target Customers

Industry	**Use Case**
Defence Shipbuilding	Design compliance & BOM synchronization
Naval Architecture	Rule validation
Aerospace	Certification compliance
Automotive	Engineering change management
Heavy Machinery	Design optimization
Oil & Gas	Standards compliance
Nuclear	Regulatory traceability
Railways	Configuration management
EPC Companies	Engineering documentation
Semiconductor Equipment	Design governance

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7. Applicability Elsewhere

Defence Shipbuilding

- Naval vessels
- Coast Guard ships
- Submarines

- Frigates
- Destroyers
- Auxiliary vessels

Commercial Shipbuilding

- Cargo ships
- LNG carriers
- Cruise ships
- Offshore support vessels
- Oil tankers

Aerospace

Applicable for:

- Aircraft design
- Structural validation
- Certification workflows
- Configuration management
- Maintenance documentation

Automotive

Supports:

- Engineering Change Management
- CAD–BOM synchronization
- Supplier validation
- Design compliance
- Manufacturing readiness

Heavy Engineering

Applicable for:

- Industrial equipment
- Turbines
- Compressors
- Mining machinery
- Process plants

Nuclear

Supports:

- Safety documentation
- Regulatory compliance
- Design verification
- Traceability

Railways

Applicable for:

- Rolling stock
- Locomotives
- Metro systems
- Maintenance engineering

Renewable Energy

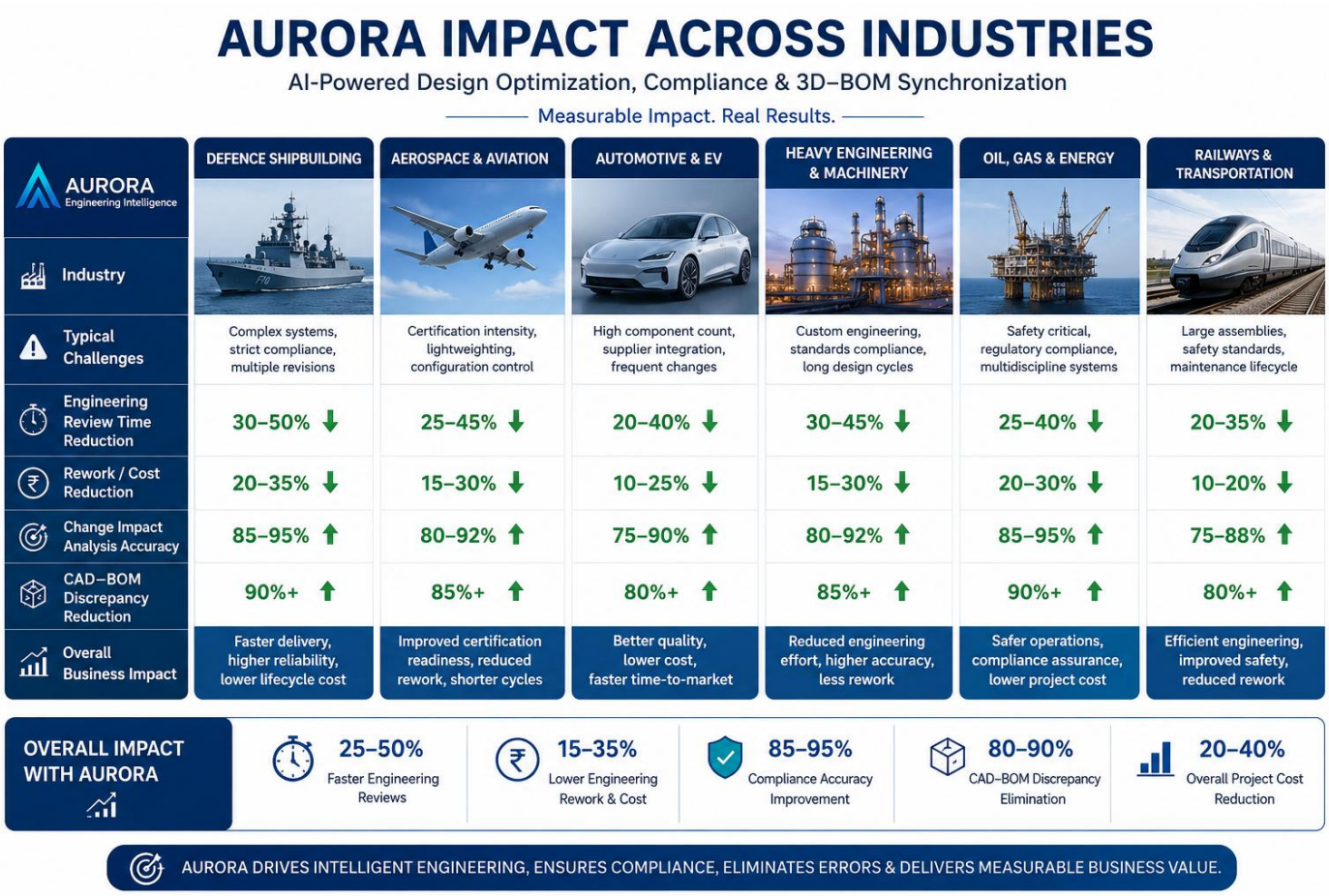
Supports:

- Wind turbines
- Solar infrastructure
- Offshore wind platforms
- Grid equipment

Smart Manufacturing

Applicable for:

- Industry 4.0
- Digital Factory
- Digital Twin
- Engineering Knowledge Management



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8. Competitive Landscape

Capability	Traditional CAD	PLM	Rule Engines	AURORA
CAD Authoring	✓	✓	✗	✓
BOM Sync	Partial	✓	✗	✓
AI Rule Compliance	✗	✗	Partial	✓
Engineering Knowledge Graph	✗	✗	✗	✓
Change Impact Prediction	✗	Partial	✗	✓
Explainable AI	✗	✗	✗	✓
Cross-disciplinary Analysis	Partial	Partial	✗	✓
Automated Recommendations	✗	✗	Partial	✓

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9. Value Proposition

AURORA delivers measurable business benefits by:

- Reducing engineering review time

- Minimizing design errors
- Eliminating CAD–BOM mismatches
- Improving first-pass design quality
- Accelerating engineering approvals
- Reducing engineering rework
- Improving compliance with standards
- Enhancing configuration management
- Increasing engineering productivity
- Providing complete engineering traceability

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10. Commercialization Roadmap

Phase	**Timeline**	**Outcome**
Prototype	Year 1	MDL Pilot
Defence Validation	Year 2	Multiple Indian Shipyards
National Expansion	Year 3	Aerospace, Railways, Heavy Engineering
International Expansion	Year 4	ASEAN, Middle East, Europe
Global Engineering Platform	Year 5	Enterprise AI Engineering Suite

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11. Conclusion

AURORA is more than an AI-assisted design verification tool; it is a comprehensive Engineering Intelligence Platform that augments existing CAD, PLM, and ERP ecosystems with explainable AI, automated compliance checking, engineering knowledge management, and intelligent change-impact analysis. Its capabilities align closely with the requirements outlined in the MDL Scope of Work, including automated compliance against class rules, downstream impact prediction, 3D–BOM synchronization, revision management, and secure on-premises deployment.

The platform is applicable across defence shipbuilding, commercial maritime, aerospace, automotive, heavy engineering, railways, energy, and other engineering-intensive sectors. By reducing manual effort, improving traceability, and enabling faster, more reliable engineering decisions, AURORA provides a scalable foundation for digital engineering transformation while supporting the broader objectives of Industry 4.0 and AI-enabled product lifecycle management.